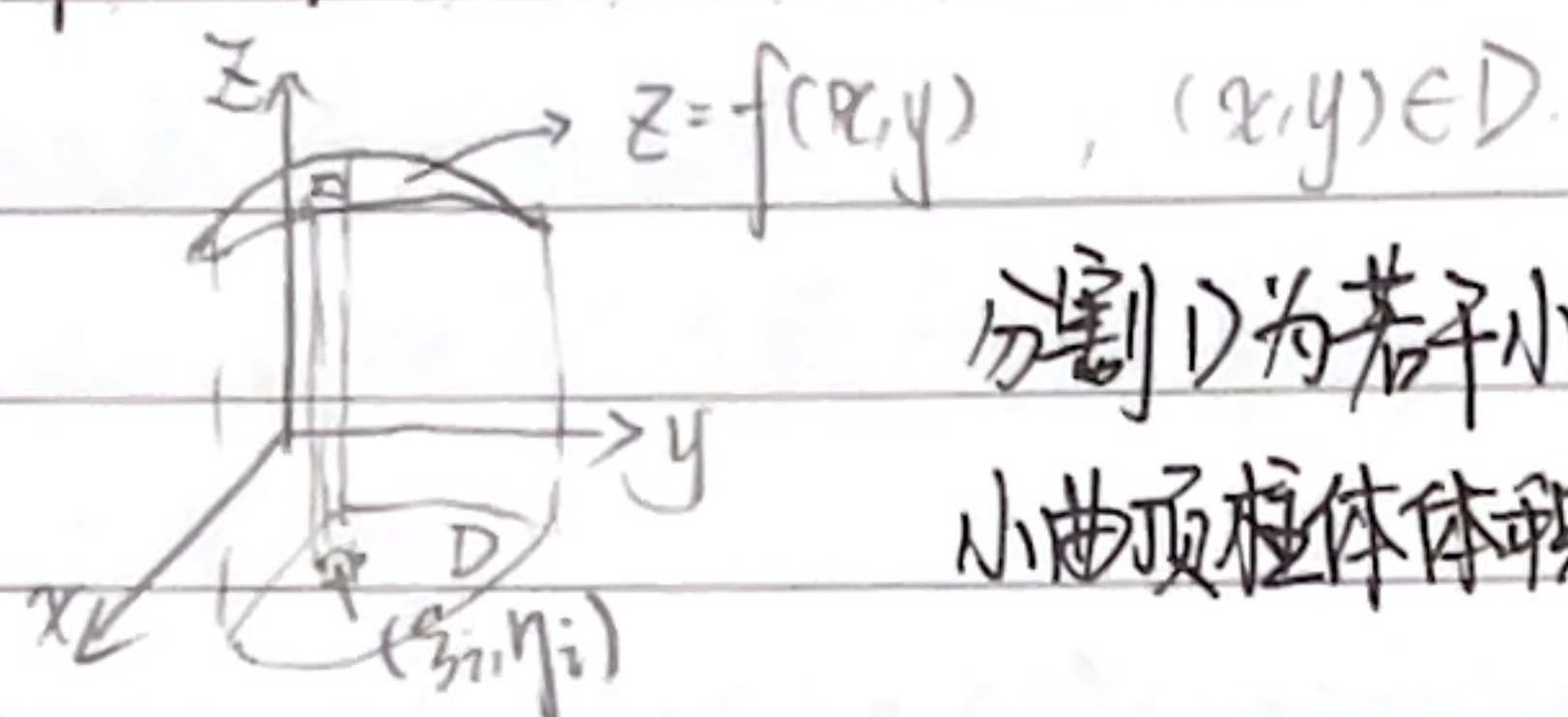


Chapter 9 二重积分

一. 概念

引例: 曲顶柱体体积

分割D为若干小区域 $D = \bigcup_{i=1}^n \Delta\sigma_i$ 小曲顶柱体体积 $\Delta V_i \approx f(\xi_i, \eta_i) \Delta\sigma_i$
小平顶体积总体积 $V = \sum_{i=1}^n \Delta V_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i) \Delta\sigma_i$ 其中 λ 是所有 $\Delta\sigma_i$ 对应图形的直径最大定义: $D \subset \mathbb{R}^2$ 为有界区域, $f(x, y)$ 在 D 上有界

$$\iint_D f(x, y) d\sigma = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i) \Delta\sigma_i$$

面积元素

注: (1)

用平行于 x, y 轴的直线划分 D , 则 $d\sigma = dxdy$

$$\iint_D f(x, y) d\sigma = \iint_D f(x, y) dxdy$$

(2) 当 $f(x, y)$ 在 D 上连续时, $\iint_D f(x, y) d\sigma$ 存在

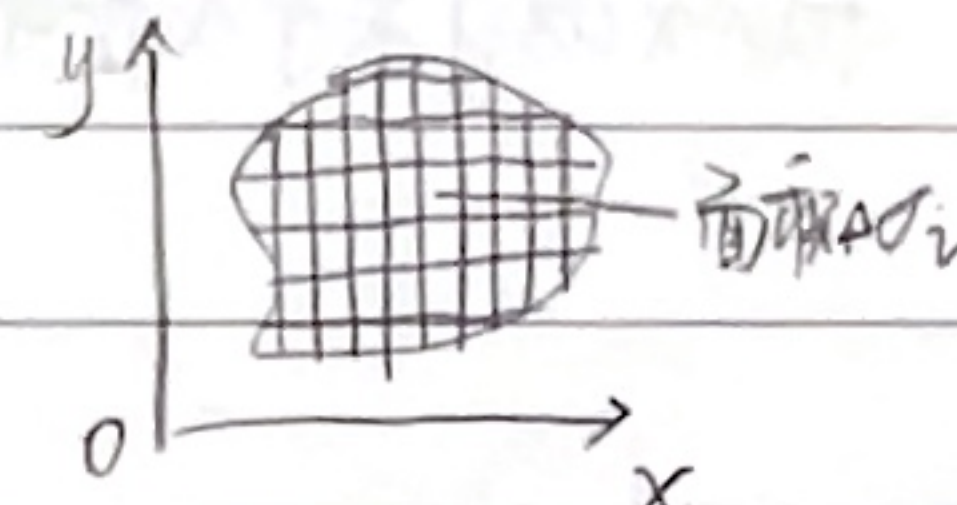
(3) 几何意义:

在 D 上:① $f(x, y) > 0$, 则 $\iint_D f(x, y) d\sigma =$ 曲顶柱体体积 ($V_{柱}$)② $f(x, y) < 0$, 则 $\iint_D f(x, y) d\sigma = - \lim_{n \rightarrow \infty} \sum_{i=1}^n (-f(\xi_i, \eta_i)) \Delta\sigma_i =$ 负曲顶柱体体积③ $f(x, y)$ 在 D 上有正有负, 则 $\iint_D f(x, y) d\sigma = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i) \Delta\sigma_i + \lim_{n \rightarrow \infty} \sum_{j=1}^m (-f(\xi_j, \eta_j)) \Delta\sigma_j$...
($D = D_+ \cup D_-$)

$$= \iint_{D_+} f(x, y) d\sigma + \iint_{D_-} f(x, y) d\sigma$$

$$= V_{柱}^{(+)} - V_{柱}^{(-)}$$

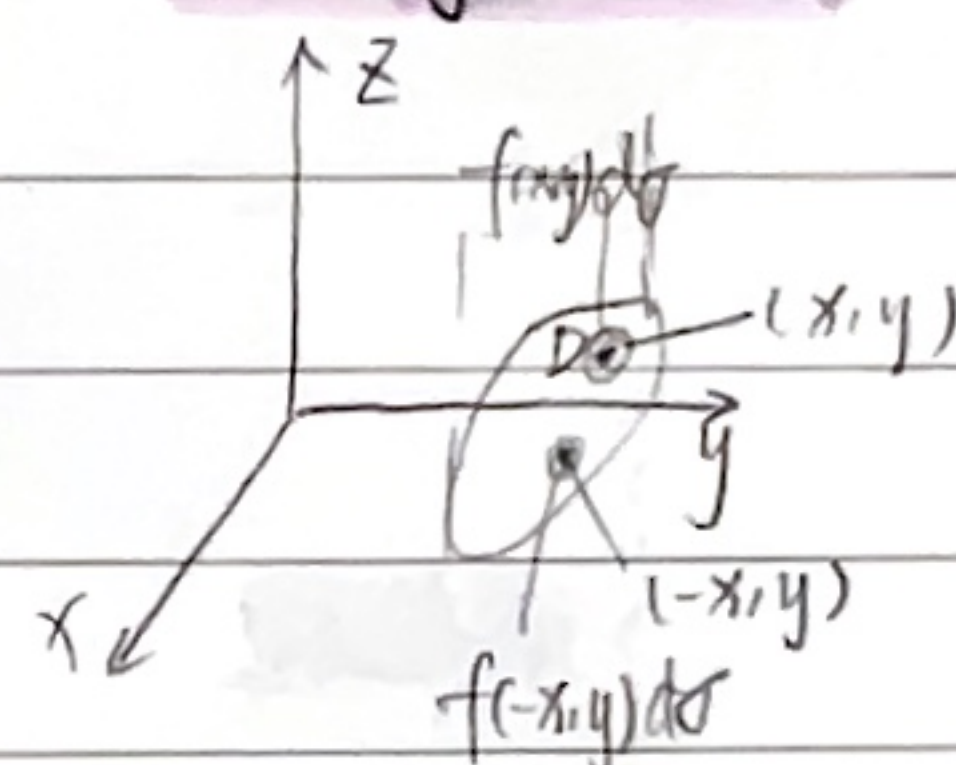
= 各不同区域曲顶柱体体积的代数和

(4) 物理意义: 平面薄板 (面密度函数 $f(x, y)$) 若 $f(x, y) > 0$ 

$$\text{总质量 } m = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i, \eta_i) \Delta\sigma_i = \iint_D f(x, y) d\sigma$$

质量元素

(5) 对称性与奇偶性

① D 关于 y 轴对称

$$\iint_D f(x, y) d\sigma = \begin{cases} 0 & f(-x, y) = -f(x, y) \quad f(x, y) \text{ 关于 } x \text{ 是奇函数} \\ 2 \iint_{D_+} f(x, y) d\sigma & f(-x, y) = f(x, y) \quad \text{偶函数} \end{cases}$$

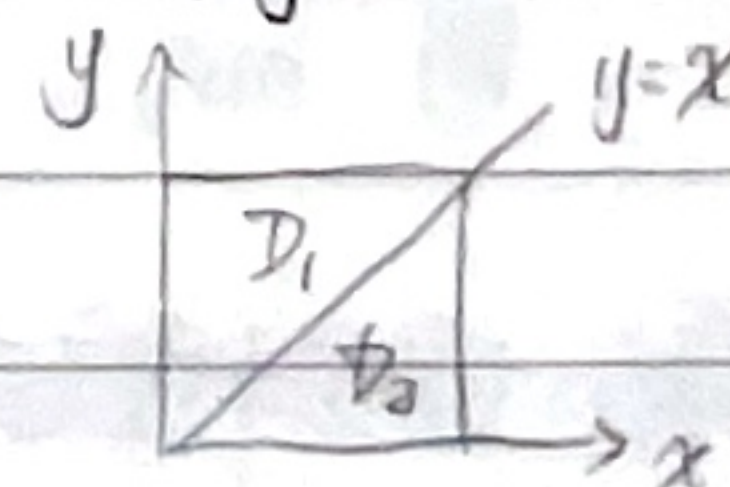
② D 关于 x 轴对称

$$\iint_D f(x, y) d\sigma = \begin{cases} 0 & f(x, -y) = -f(x, y) \quad f(x, y) \text{ 关于 } y \text{ 是奇函数} \\ 2 \iint_{D_+} f(x, y) d\sigma & f(x, -y) = f(x, y) \quad y \text{ 是偶} \end{cases}$$

③ D 关于原点对称 $D = D_1 \cup D_2$

$$\iint_D f(x, y) d\sigma = \begin{cases} 0 & f(-x, -y) = -f(x, y) \\ 2 \iint_{D_1} f(x, y) d\sigma & f(-x, -y) = f(x, y) \end{cases}$$

④ 轮换对称性

若 $(x, y) \in D$ 且 $(y, x) \in D$ 则区域 D 具有轮换性. $\Leftrightarrow D$ 关于 $y=x$ 对称. $\rightarrow$ 若 D 具有轮换性, 则 $\iint_D f(x, y) d\sigma = \iint_D f(y, x) d\sigma$ ⑤ $\iint_D d\sigma = \iint_D 1 d\sigma = D$ 的面积

高度为1的平顶柱体的体积

Somewhere, something incredible

is waiting to be found.

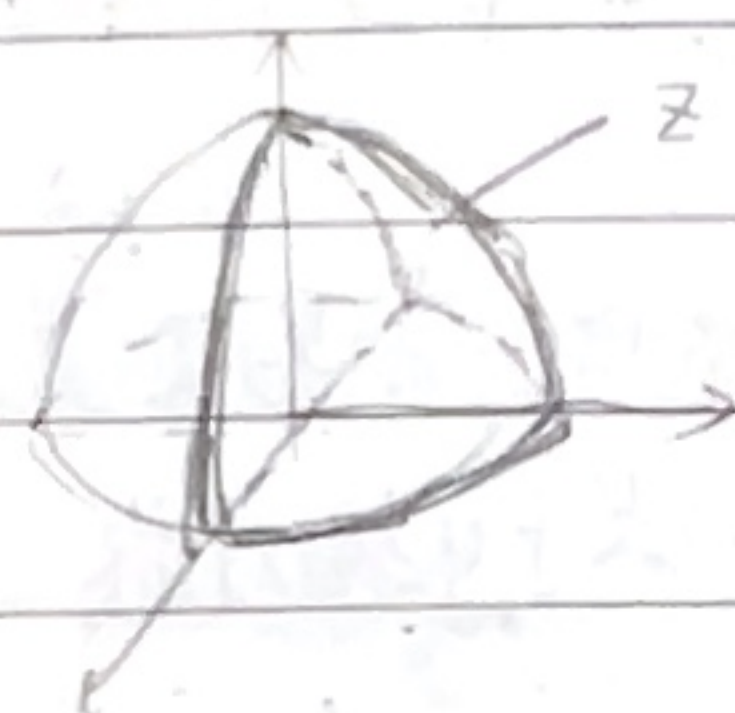
[例1] (1) $I = \iint_D (1 + \tan^2 x \cos y) d\sigma$, 其中 $D: x^2 + y^2 \leq 4, y \geq 0$.

D 关于 y 轴对称
 $\tan^2 x \cos y$ 关于 x 是奇函数

$$I = \iint_D 1 d\sigma + \iint_D \tan^2 x \cos y d\sigma = \frac{1}{2} \pi \cdot 4 + 0 = 2\pi$$

(2) $\iint_D \sqrt{4-x^2-y^2} d\sigma$, $D: x^2 + y^2 \leq 4, y \geq 0$

$$V_{\text{曲顶}} = \frac{1}{4} V_{\text{球}} = \frac{1}{4} \cdot \frac{4}{3} \pi \cdot 2^3 = \frac{8}{3} \pi.$$



二、二重积分的性质

性质1: $\iint_D k f(x, y) d\sigma = k \iint_D f(x, y) d\sigma \quad (k \in \mathbb{R})$

线性性

$$\iint_D (\alpha f(x, y) + \beta g(x, y)) d\sigma = \alpha \iint_D f(x, y) d\sigma + \beta \iint_D g(x, y) d\sigma$$

性质2: (区域可加性) $D = D_1 \cup D_2$, D_1, D_2 无公共区域.

$$\iint_D f(x, y) d\sigma = \iint_{D_1} f(x, y) d\sigma + \iint_{D_2} f(x, y) d\sigma$$

性质3: (保序性) 若 $f(x, y) \leq g(x, y)$, $(x, y) \in D$, 则

$$\iint_D f(x, y) d\sigma \leq \iint_D g(x, y) d\sigma$$

性质4: (估值结果) $m \leq f(x, y) \leq M, (x, y) \in D$

$$\text{则 } m\sigma_D \leq \iint_D f(x, y) d\sigma \leq M\sigma_D, \text{ 其中 } \sigma_D \text{ 为 } D \text{ 的面积}$$

性质5: (积分中值定理) $f(x, y)$ 在闭区域 D 上连续, 则

$$\iint_D f(x, y) d\sigma = f(\xi, \eta) \sigma_D, \text{ 其中 } (\xi, \eta) \in D.$$

$$m \leq \frac{1}{\sigma_D} \iint_D f(x, y) d\sigma \leq M \quad \Rightarrow \quad f(\xi, \eta) = \frac{1}{\sigma_D} \iint_D f(x, y) d\sigma$$

介值定理

[例2] $D: x^2 + y^2 \leq 4$. 估算 $\iint_D (x^2 + 4y^2 + 1) d\sigma$.

解: 令 $x = r \cos \theta, y = r \sin \theta, 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi$

$$f(x, y) = x^2 + 4y^2 + 1 = r^2 \cos^2 \theta + 4r^2 \sin^2 \theta + 1 = r^2 + 1 + 3r^2 \sin^2 \theta$$

$$f(r, \theta)_{\max} = 4 + 1 + 3 \times 4 = 17$$

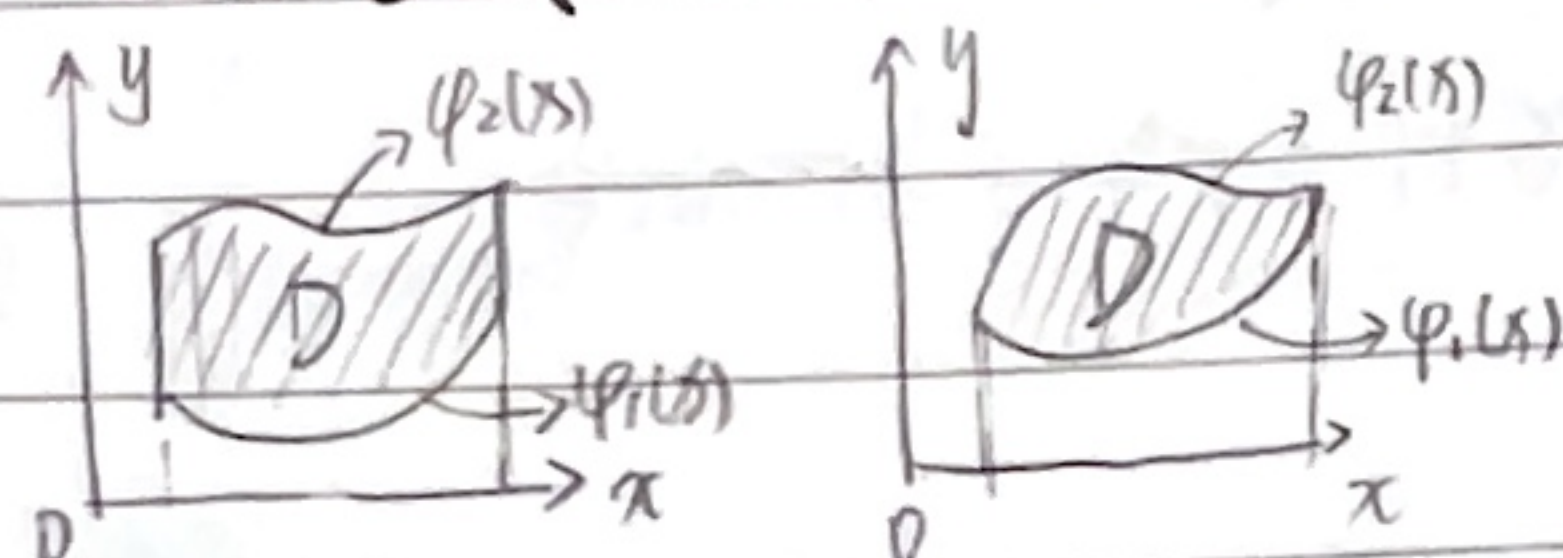
$$f(r, \theta)_{\min} = 0 + 1 + 0 = 1 \quad \text{又 } \sigma_D = 4\pi$$

$$4\pi \leq \iint_D (x^2 + 4y^2 + 1) d\sigma \leq 68\pi$$

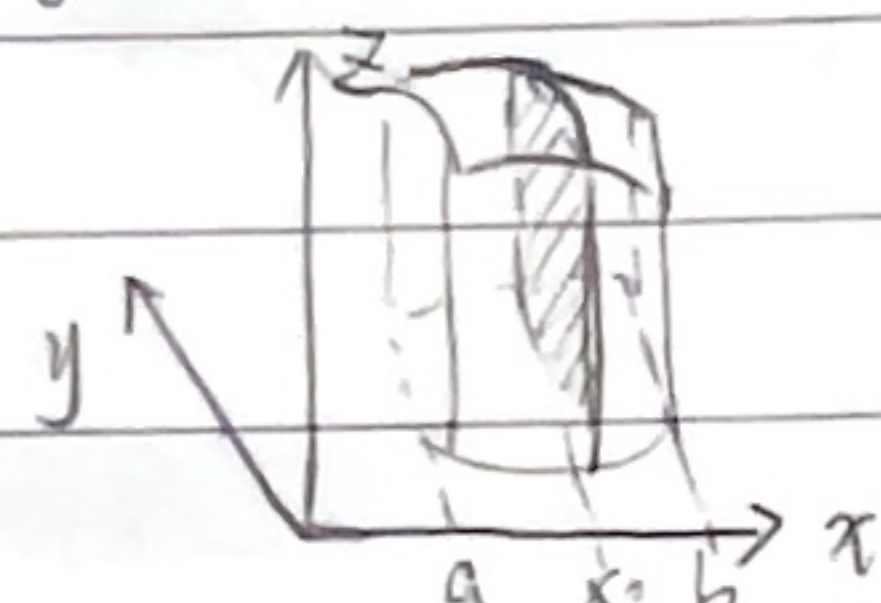
§2. 二重积分的计算

一. 直角坐标系

(一) 垂直型区域 (x型)



$$D: a \leq x \leq b, \phi_1(x) \leq y \leq \phi_2(x)$$

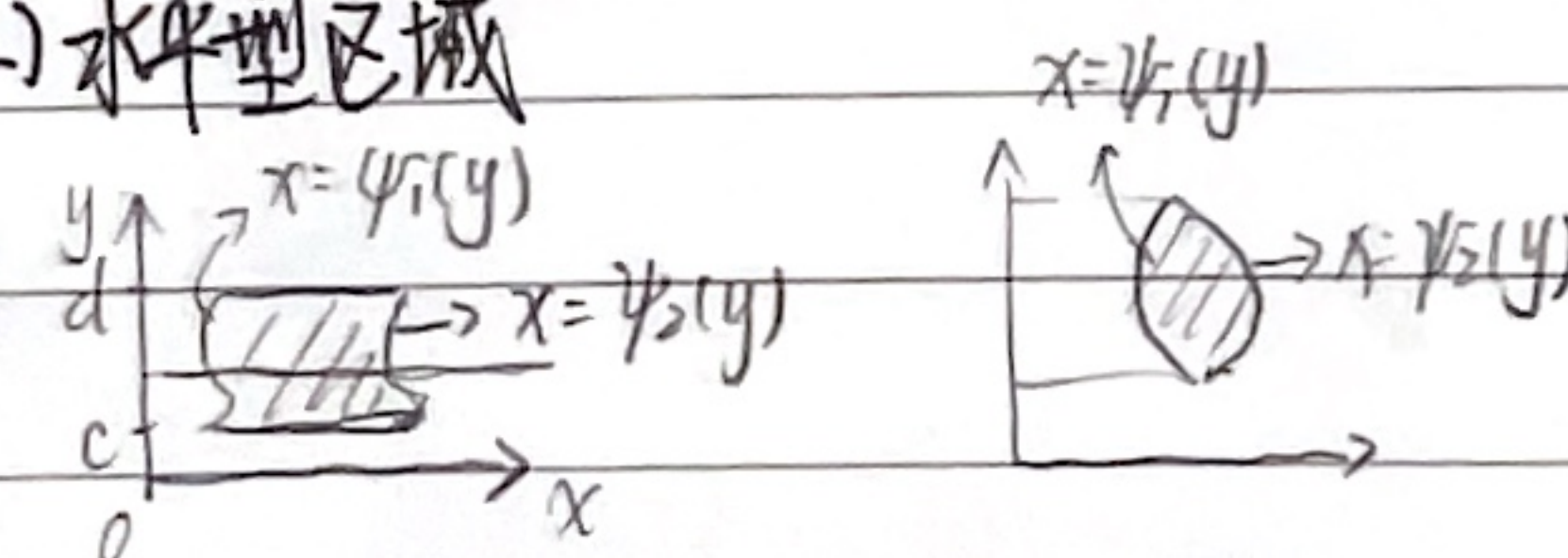


由截面面积已知的立体体积/式

$$V = \int_a^b A(x) dx = \int_a^b \left(\int_{\phi_1(x)}^{\phi_2(x)} f(x,y) dy \right) dx$$

$$\iint_D f(x,y) d\sigma = \int_a^b dx \int_{\phi_1(x)}^{\phi_2(x)} f(x,y) dy$$

(二) 水平型区域



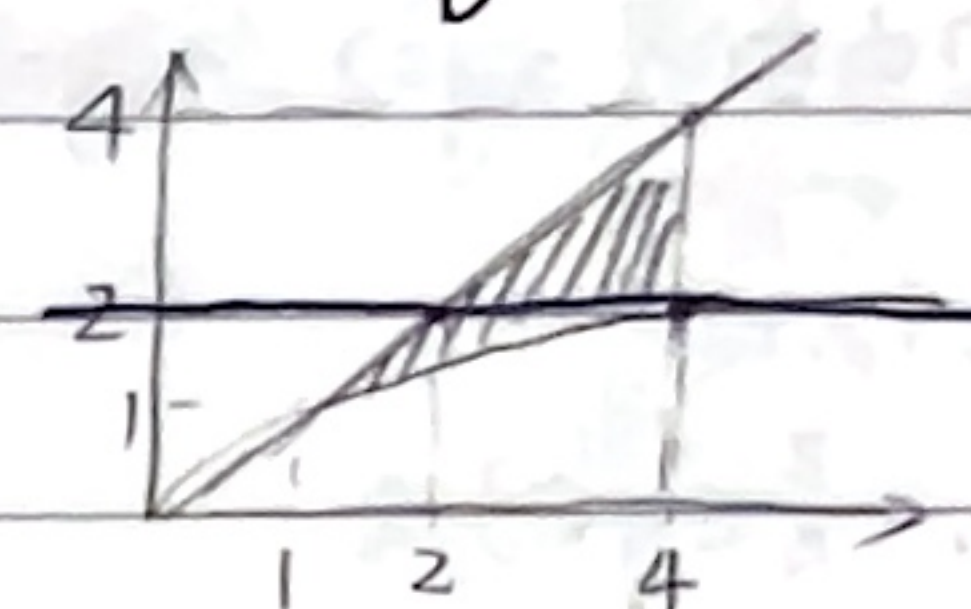
$$D: c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)$$

$$\iint_D f(x,y) dx = \int_c^d dy \int_{\psi_1(y)}^{\psi_2(y)} f(x,y) dx$$

[例1] 设D由 $\sqrt{x} \leq y \leq x$, $x \leq 4$ 确定, 用两种不同次序表示 $\iint_D f(x,y) d\sigma$

$$D: 1 \leq x \leq 4, \sqrt{x} \leq y \leq x$$

$$\therefore \iint_D f(x,y) d\sigma = \int_1^4 dx \int_{\sqrt{x}}^x f(x,y) dy$$



$$D = D_1 + D_2$$

$$D_1: 2 \leq y \leq 4, y \leq x \leq 4$$

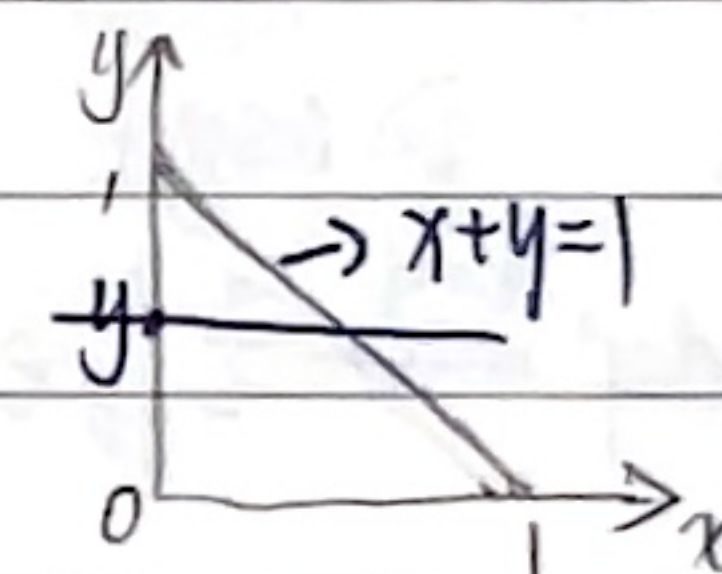
$$D_2: 1 \leq y \leq 2, y \leq x \leq y^2$$

$$\therefore \iint_D f(x,y) d\sigma = \int_2^4 dy \int_y^4 f(x,y) dx + \int_1^2 dy \int_{y^2}^4 f(x,y) dx$$

[例2] 交换积分次序

$$(1) \int_0^1 dx \int_0^{1-x} f(x,y) dy$$

$$D: 0 \leq x \leq 1, 0 \leq y \leq 1-x$$



$$D: 0 \leq y \leq 1, 0 \leq x \leq 1-y$$

$$I = \int_0^1 dy \int_0^{1-y} f(x,y) dx$$

$$(2) \int_0^1 dx \int_0^{\sqrt{2x-x^2}} f(x,y) dy + \int_1^2 dx \int_0^{2-x} f(x,y) dy$$

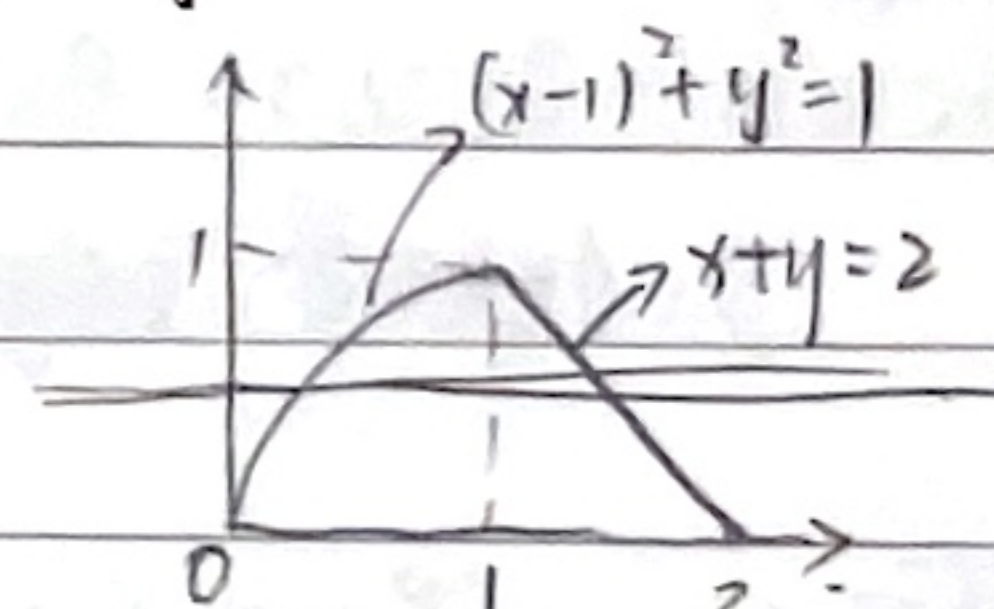
$$D = D_1 + D_2$$

$$D_1: 0 \leq x \leq 1, 0 \leq y \leq \sqrt{2x-x^2}$$

$$D_2: 1 \leq x \leq 2, 0 \leq y \leq 2-x$$

$$D: 0 \leq y \leq 1, 1-\sqrt{1-y^2} \leq x \leq 2-y$$

$$I = \int_0^1 dy \int_{1-\sqrt{1-y^2}}^{2-y} f(x,y) dx$$



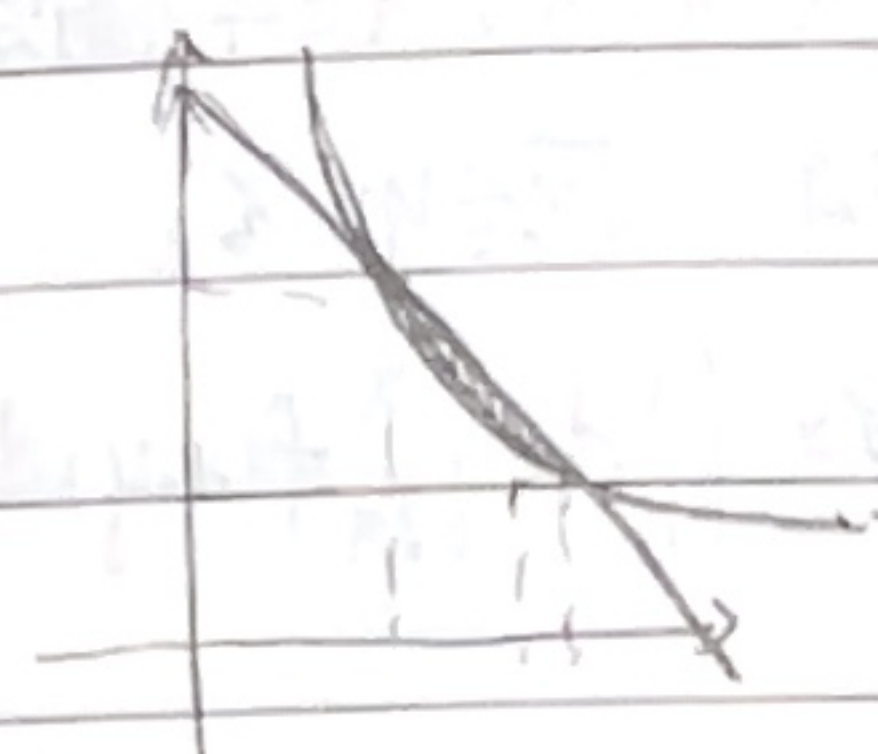
Somewhere, something incredible,

is waiting to be found.

[例3] 计算积分

(1) $\iint_D xy d\sigma$, 其中 D 由曲线 $xy=2$, $x+y=3$ 围成解: 交点 $(1,2), (2,1)$

$$D: 1 \leq x \leq 2, \frac{2}{x} \leq y \leq 3-x$$



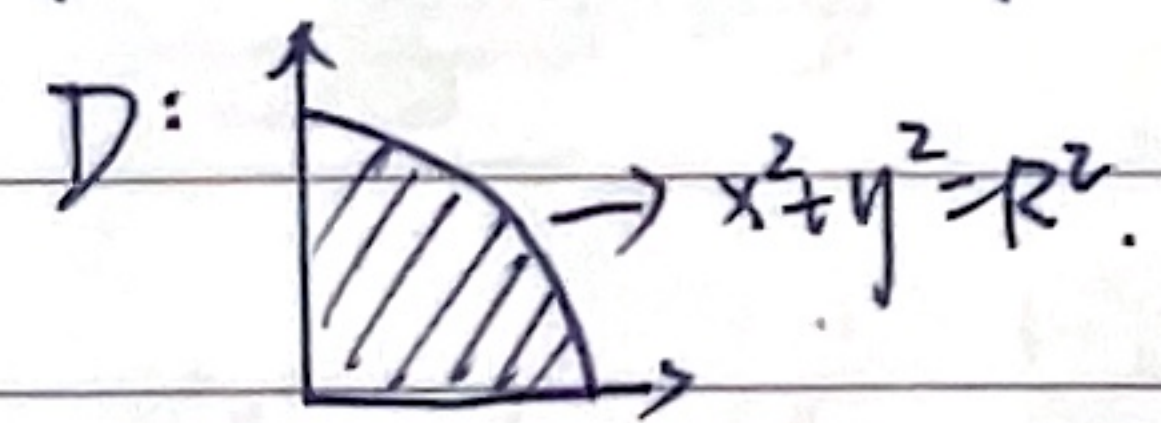
$$I = \int_1^2 dx \int_{\frac{2}{x}}^{3-x} xy dy = \int_1^2 x \cdot \frac{1}{2} [(3-x)^2 - \frac{4}{x^2}] dx = \frac{1}{2} \int_1^2 (9x - 6x^2 + x^3 - \frac{4}{x}) dx = \frac{13}{8} - \ln 4$$

(2) $\iint_D x^2 e^{-y^2} d\sigma$ D 以 $(0,0), (0,1), (1,1)$ 为顶点的三角形区域.

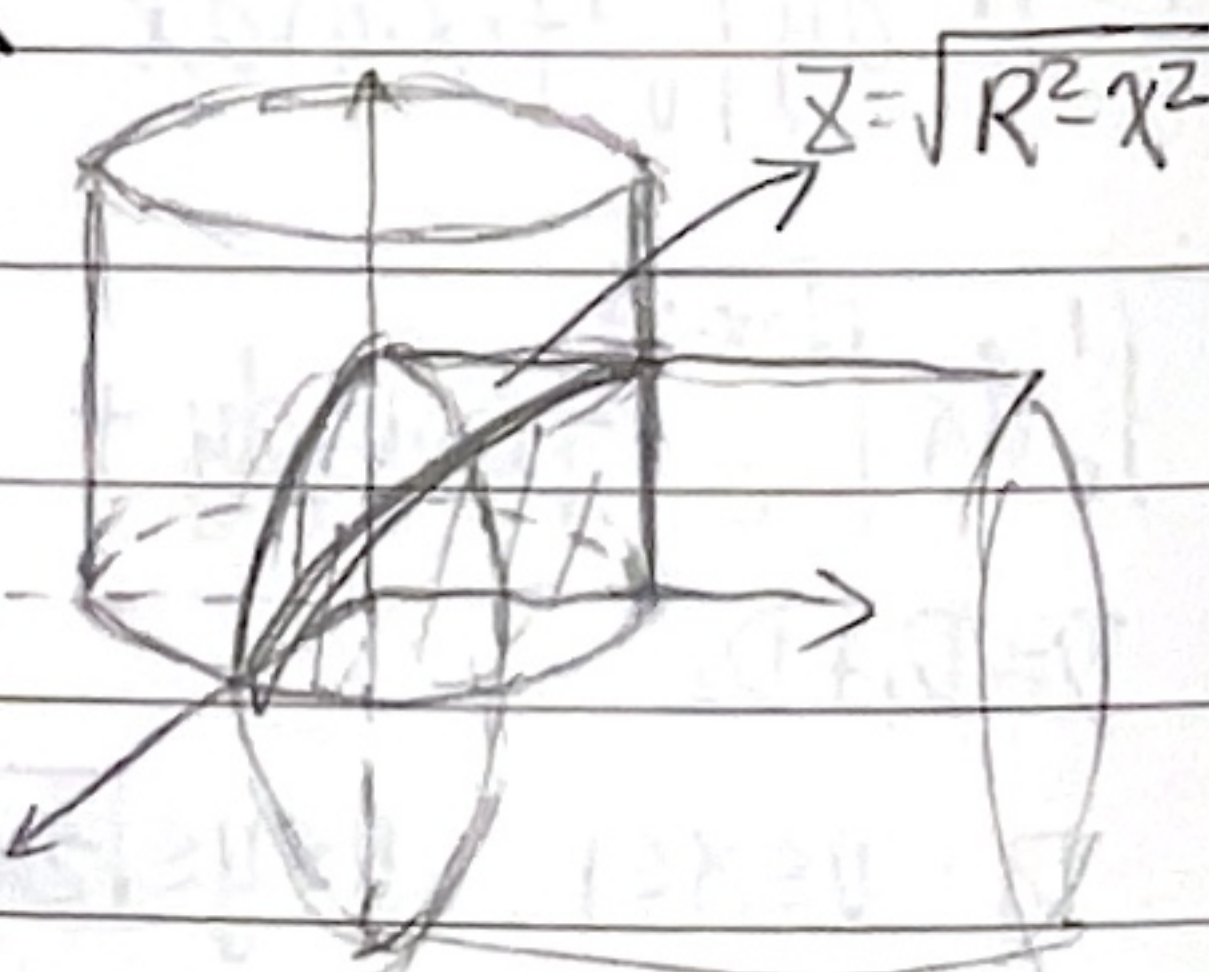
$$\int x^2 dx \int e^{-y^2} dy \quad \text{无法用初等函数表示}$$

故需考虑特定次序 $\int e^{-y^2} dy \int x^2 dx$ $D: 0 \leq y \leq 1, 0 \leq x \leq y$ $\therefore I = \int_0^1 e^{-y^2} dy \int_0^y x^2 dx$

$$I = \int_0^1 e^{-y^2} \cdot \frac{y^3}{3} dy = \frac{1}{3} \int_0^1 e^{-y^2} y^2 \cdot \frac{1}{2} dy^2 \stackrel{t=y^2}{=} \frac{1}{6} \int_0^1 t e^{-t} dt = -\frac{1}{6} (te^{-t} + e^{-t}) \Big|_0^1 = -\frac{1}{6} (1 - \frac{2}{e})$$

[例4] 求两直交圆柱 $x^2+y^2=R^2, x^2+z^2=R^2$ 所围立体体积第一象限所围曲顶柱体在 xOy 面投影

$$V_1 = \iint_D \sqrt{R^2-x^2} dx dy \quad D: 0 \leq x \leq R, 0 \leq y \leq \sqrt{R^2-x^2}$$

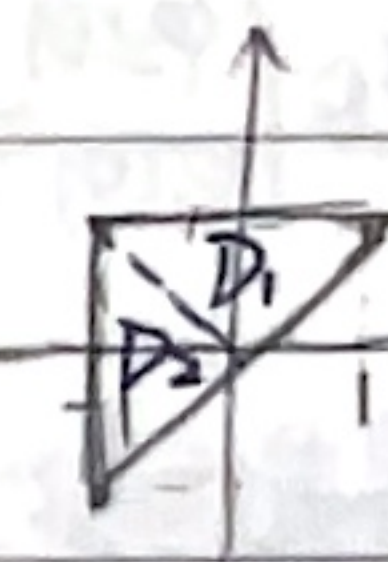


$$= \int_0^R \sqrt{R^2-x^2} dx \int_0^{\sqrt{R^2-x^2}} dy = \int_0^R (R^2-x^2) dx = \frac{2}{3} R^3$$

$$V = 8V_1 = \frac{16}{3} R^3$$

[例5] D 是以 $(1,1), (-1,1), (-1,-1)$ 为顶点的三角形区域, 求 $\iint_D (xy + \cos x \sin y) d\sigma$ $D = D_1 + D_2$ D_1 关于 y 轴对称, D_2 关于 x 轴对称.

$$\iint_{D_1} xy d\sigma = 0 \quad \iint_{D_1} \cos x \sin y d\sigma = 2 \iint_{D_1} \cos x \sin y d\sigma \quad \text{关于 } x \text{ 偶.}$$

 $D_1, D_2:$

$$0 \leq y \leq 1$$

$$0 \leq x \leq y$$

$$\iint_{D_2} (xy + \cos x \sin y) d\sigma = 0$$

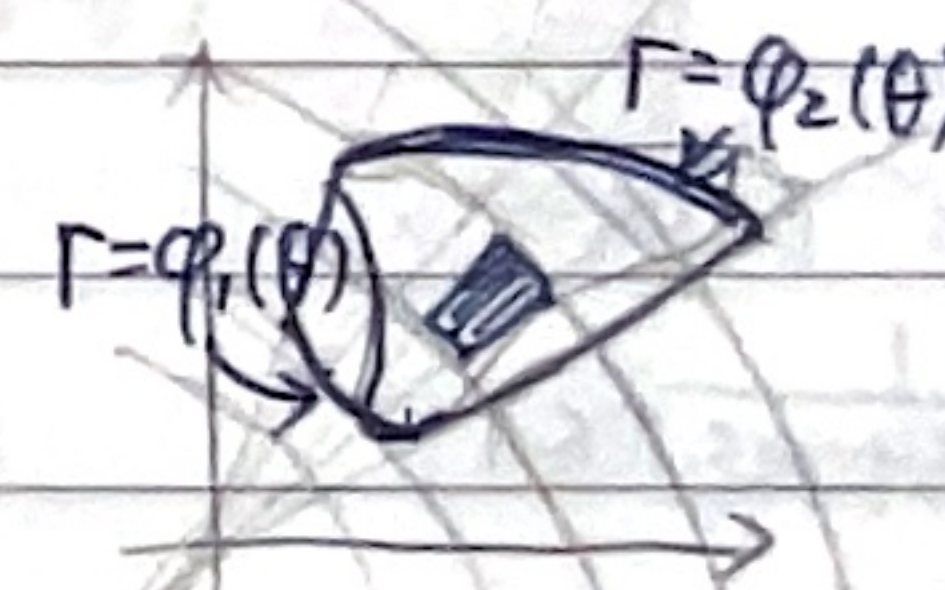
$$I = \iint_D (xy + \cos x \sin y) d\sigma = 2 \iint_{D_1} \cos x \sin y d\sigma = 2 \int_0^1 \sin y dy \int_0^y \cos x dx = 2 \int_0^1 \sin y dy$$

$$= 2 \int_0^1 \frac{1 - \cos 2y}{2} dy = (y - \frac{1}{2} \sin 2y) \Big|_0^1 = 1 - \frac{1}{2} \sin 2$$

二. 极坐标系

$$\iint_D f(x,y) d\sigma \quad (r, \theta) \quad 0 \leq r < +\infty, 0 \leq \theta < 2\pi$$

$$\text{坐标线} \begin{cases} x = \text{常数} \\ y = \text{常数} \end{cases} \rightarrow \begin{cases} r = \text{常数} \\ \theta = \text{常数} \end{cases}$$



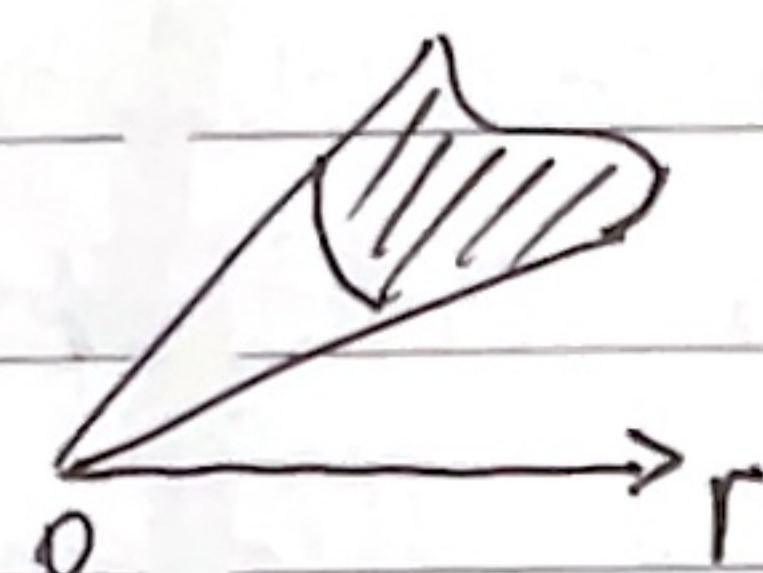
$$\text{小区域圆形面积} \Delta \sigma \approx \frac{1}{2} (r + \Delta r)^2 \Delta \theta - \frac{1}{2} r^2 \Delta \theta = r \Delta r \Delta \theta + o(\Delta r) \Delta \theta$$

面积元素 $d\sigma = r dr d\theta$ (用长方形面积近似长 r , 宽 $r d\theta$)

$$\iint_D f(x,y) d\sigma = \iint_D f(r \cos \theta, r \sin \theta) r dr d\theta$$

(1) $D: \alpha \leq \theta \leq \beta$, $\varphi_1(\theta) \leq r \leq \varphi_2(\theta)$ 且极点 $O \notin D$.

$$\iint_D f(x, y) dr = \int_{\alpha}^{\beta} d\theta \int_{\varphi_1(\theta)}^{\varphi_2(\theta)} f(r \cos \theta, r \sin \theta) r dr$$



(2) 极点 O 在区域 D 边界或内部

$$D: \alpha \leq \theta \leq \beta, 0 \leq r \leq r(\theta)$$

$$\iint_D f(x, y) dr = \int_{\alpha}^{\beta} d\theta \int_0^{r(\theta)} f(r \cos \theta, r \sin \theta) r dr$$

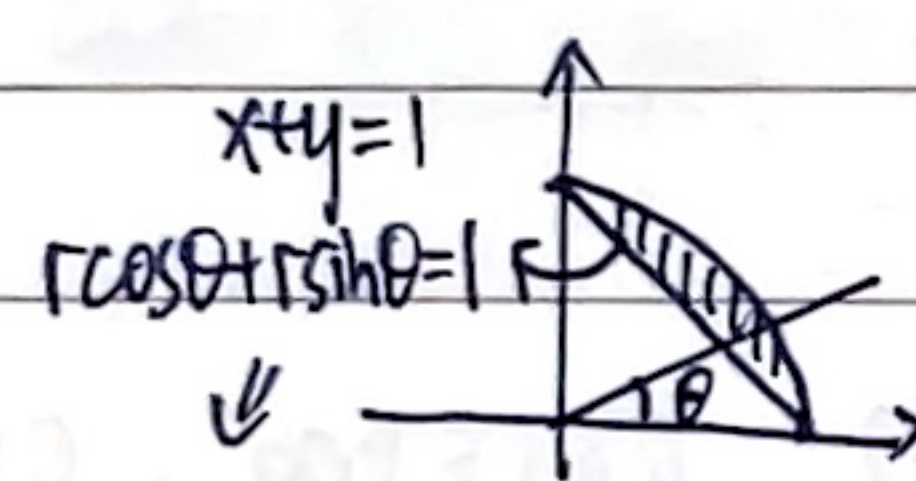


$$\iint_D f(x, y) dr = \int_0^{2\pi} d\theta \int_0^{r(\theta)} f(r \cos \theta, r \sin \theta) r dr$$



[例1] 直角坐标和极坐标互化

(1) $\iint_D f(x, y) dr$, $D: \begin{cases} x+y \geq 1 \\ y \leq \sqrt{1-x^2} \end{cases}$



$$r = \frac{1}{\sin \theta + \cos \theta}$$

$$D: 0 \leq \theta \leq \frac{\pi}{2}$$

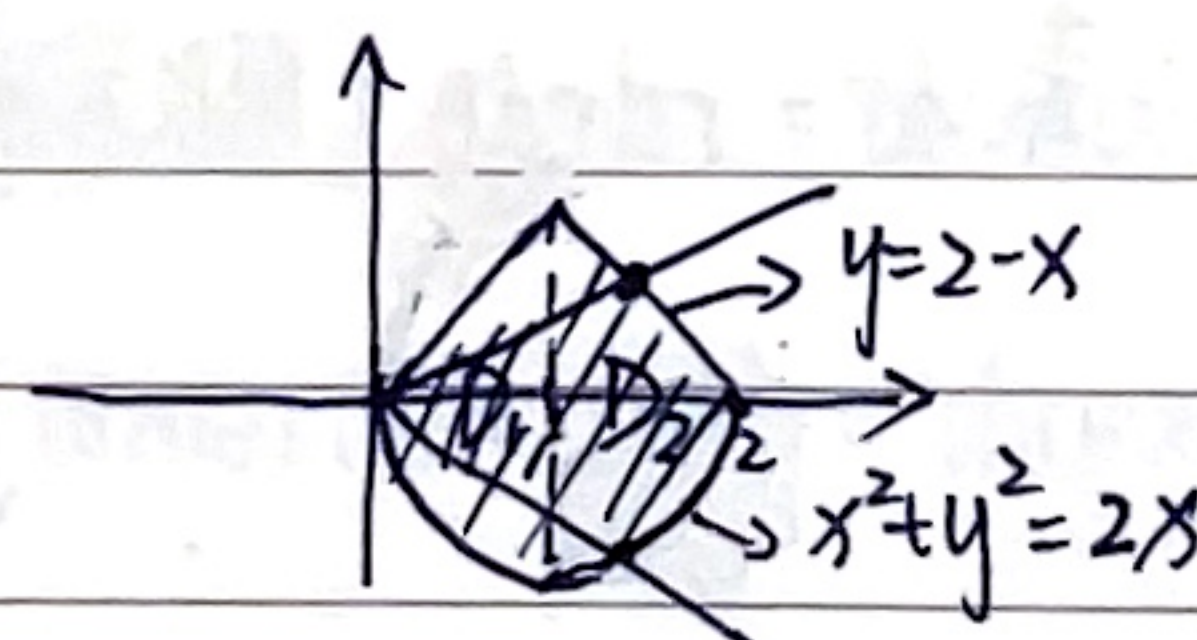
$$\frac{1}{\sin \theta + \cos \theta} \leq r \leq 1$$

$$\therefore I = \int_0^{\frac{\pi}{2}} d\theta \int_{\frac{1}{\sin \theta + \cos \theta}}^1 f(r \cos \theta, r \sin \theta) r dr$$

(2) $\int_0^1 dx \int_{-\sqrt{2x-x^2}}^x f(x, y) dy + \int_1^2 dx \int_{-\sqrt{2x-x^2}}^{2-x} f(x, y) dy$

$$D = D_1 \cup D_2 \quad D_1: \begin{cases} 0 \leq \theta \leq \frac{\pi}{4} \\ 0 \leq r \leq \frac{2}{\sin \theta + \cos \theta} \end{cases}$$

$$D_2: \begin{cases} -\frac{\pi}{2} \leq \theta \leq 0 \\ 0 \leq r \leq 2 \cos \theta \end{cases}$$



$$I = \int_{-\frac{\pi}{2}}^0 d\theta \int_0^{2 \cos \theta} f(r \cos \theta, r \sin \theta) r dr + \int_0^{\frac{\pi}{4}} d\theta \int_{\frac{2}{\sin \theta + \cos \theta}}^2 f(r \cos \theta, r \sin \theta) r dr$$

[例2] 计算:

(1) $\iint_D e^{-x^2-y^2} dr$, $D: x^2+y^2 \leq 1$.

$$D: \begin{cases} 0 \leq \theta \leq 2\pi \\ 0 \leq r \leq 1 \end{cases} \quad I = \int_0^{2\pi} d\theta \int_0^1 e^{-r^2} r dr = -\frac{1}{2} \int_0^{2\pi} (e^{-1} - 1) d\theta = -\pi(e^{-1} - 1)$$

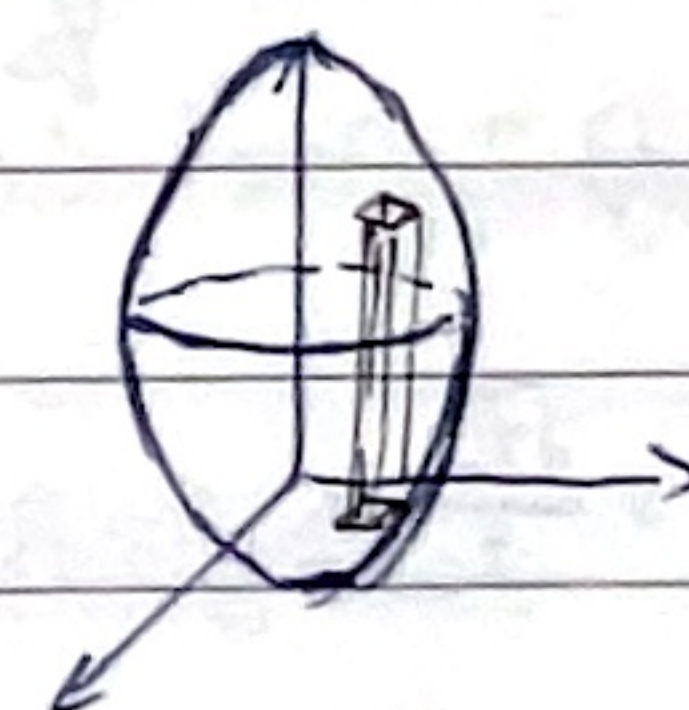
(2) $\iint_D |x^2+y^2-z| dr$, $D: x^2+y^2 \leq 3$

$$\begin{aligned} \iint_{x^2+y^2 \leq 2} (2-x^2-y^2) dr + \iint_{2 \leq x^2+y^2 \leq 3} (x^2+y^2-z) dr &= \int_0^{2\pi} d\theta \int_0^{\sqrt{2}} (2-r^2) r dr + \int_0^{2\pi} d\theta \int_{\sqrt{2}}^{\sqrt{3}} (r^2-2) r dr \\ &= 2\pi \cdot \left[r^2 - \frac{1}{4} r^4 \right]_0^{\sqrt{2}} + 2\pi \cdot \left[\frac{1}{4} r^4 - r^2 \right]_{\sqrt{2}}^{\sqrt{3}} = \frac{5}{2} \pi \end{aligned}$$

(3) 求曲面 $z=5-x^2-y^2$ 及 $z=x^2+y^2-3$ 所围立体体积

$\rightarrow z=5-y^2$ 绕 z 轴 $\rightarrow z=y^2-3$ 绕 z 轴.

解: 曲面交线 $\Rightarrow \begin{cases} z=1 \\ x^2+y^2=4 \end{cases}$



$\therefore$ 立体在 xOy 平面投影区域 $D_{xy}: x^2+y^2 \leq 4$

$$V = \iint_{D_{xy}} (z_2 - z_1) dr = \iint_{D_{xy}} [5 - x^2 - y^2 - (x^2 + y^2 - 3)] dr = \iint_{D_{xy}} (8 - 2x^2 - 2y^2) dr$$

$$= \int_0^{2\pi} d\theta \int_0^2 (8 - 2r^2) r dr = 16\pi.$$

$$(4) \iint_D \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) d\sigma, D: x^2 + y^2 \leq R^2$$

$$\because D \text{ 关于 } y=x \text{ 对称} \therefore \iint_D x^2 d\sigma = \iint_D y^2 d\sigma = \frac{1}{2} \iint_D (x^2 + y^2) d\sigma$$

$$\therefore \text{原式} = \frac{1}{a^2} \iint_D x^2 d\sigma + \frac{1}{b^2} \iint_D y^2 d\sigma = \frac{1}{2a^2} \iint_D (x^2 + y^2) d\sigma + \frac{1}{2b^2} \iint_D (x^2 + y^2) d\sigma$$

$$= \left(\frac{1}{2a^2} + \frac{1}{2b^2} \right) \int_0^{2\pi} d\theta \int_0^R r^2 \cdot r dr = \frac{1}{4} \left(\frac{1}{a^2} + \frac{1}{b^2} \right) \pi R^4$$

(5) 计算概率积分

$$I = \int_0^{+\infty} e^{-x^2} dx = \lim_{R \rightarrow +\infty} I(R) \quad \text{其中 } I(R) = \int_0^R e^{-x^2} dx$$

$$I^2(R) = \int_0^R e^{-x^2} dx \cdot \int_0^R e^{-y^2} dy = \iint_{D_1} e^{-x^2-y^2} d\sigma \quad D_1: [0, R] \times [0, R]$$

$$\iint_{D_1} e^{-x^2-y^2} d\sigma \leq I^2(R) \leq \iint_{D_2} e^{-x^2-y^2} d\sigma \quad \iint_{D_2} e^{-x^2-y^2} d\sigma = \frac{\pi}{4} (e^{-2R^2} - 1) \rightarrow \frac{\pi}{4} (R \rightarrow +\infty)$$

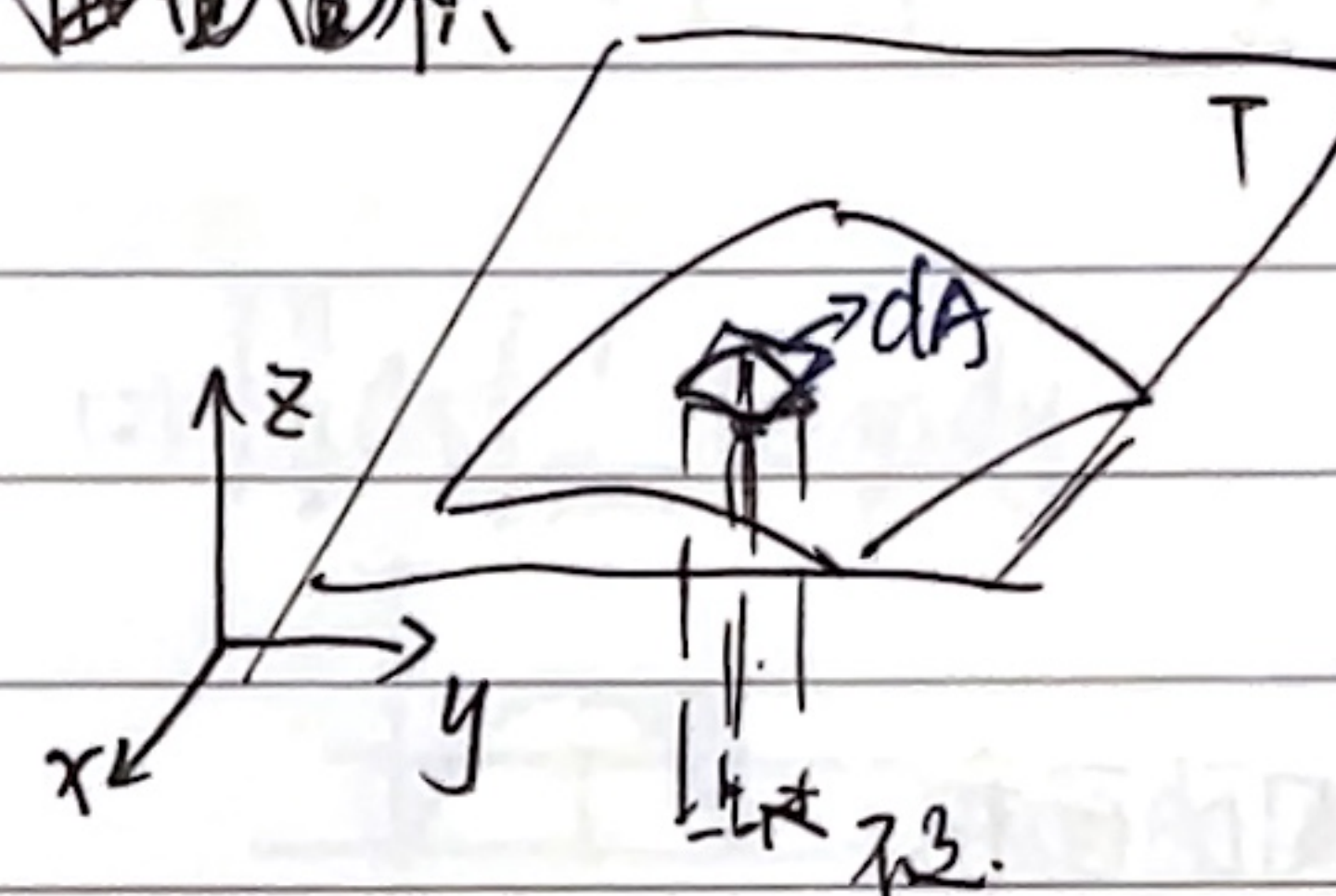
$$\text{同理 } \iint_{D_1} e^{-x^2-y^2} d\sigma = \frac{\pi}{4}$$

$$\Rightarrow I = \frac{\sqrt{\pi}}{2}$$

§4. 重积分的应用.

$$\iint_D f(x, y) d\sigma \begin{cases} \text{曲顶柱体的体积} \\ D \text{ 的面积 (} f(x, y) = 1 \text{)} \\ \text{平面薄板质量} \end{cases}$$

曲面面积



$$\frac{d\sigma}{dA} = |\cos(\widehat{xOy \text{ 面}, \text{切平面}})|$$

$$= |\cos((0, 0, 1), \vec{n})|$$

$$\vec{n} = (-f_x, -f_y, 1)$$

$$= \frac{1}{\sqrt{1+f_x^2+f_y^2}}$$

$$\therefore dA = \sqrt{1+f_x^2+f_y^2} d\sigma$$

$$\therefore \text{曲面面积 } A = \iint_{D_{xy}} dA = \iint_{D_{xy}} \sqrt{1+f_x^2+f_y^2} d\sigma. \quad \text{其中 } D_{xy} \text{ 是曲面 } S \text{ 在 } xOy \text{ 面的投影域.}$$

注: (1) 求曲面 $z = f(x, y)$ 的面积:

项来投影, 不一定是在 xOy 面上

$$\begin{cases} \text{一投: 求 } S \text{ 在 } xOy \text{ 面投影域 } D_{xy} \\ \text{二代: 求 } dA = \sqrt{1+z_x^2+z_y^2} d\sigma \\ \text{三换: } A = \iint_{D_{xy}} \sqrt{1+z_x^2+z_y^2} d\sigma \end{cases} \begin{cases} \frac{R d\sigma}{\sqrt{R^2-x^2-y^2}} & z = \sqrt{R^2-x^2-y^2} \text{ (球面)} \\ \sqrt{z} d\sigma & z = \sqrt{x^2+y^2} \text{ (圆锥面)} \end{cases}$$

(2) 曲面方程为一般形式 $S: F(x, y, z) = 0$

① 选择合适的坐标面投影, 使投影域简单.

② 如投影到 xOz 面, D_{xz} , 视曲面为 $y = y(x, z)$, $dA = \sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2} dx dz$

$$A = \iint_{D_{xz}} \sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2} dx dz.$$

Somewhere, something incredible,

is waiting to be found.

→ $z=y$ 绕 z 旋转

[例1] 求锥面 $z=\sqrt{x^2+y^2}$ 含在平面 $z=0, z=1$ 间的面积.

解: $D_{xy}: x^2+y^2 \leq 1 \quad \therefore A = \iint_{D_{xy}} \sqrt{1+z_x^2+z_y^2} dx dy = \iint_{D_{xy}} \sqrt{2} dx dy = \sqrt{2} \pi \cdot 1^2 = \sqrt{2} \pi$

[例2] 求锥面 $z=\sqrt{x^2+y^2}$ 含在柱体 $x^2+y^2 \leq 2x$ 内那一部分面积.

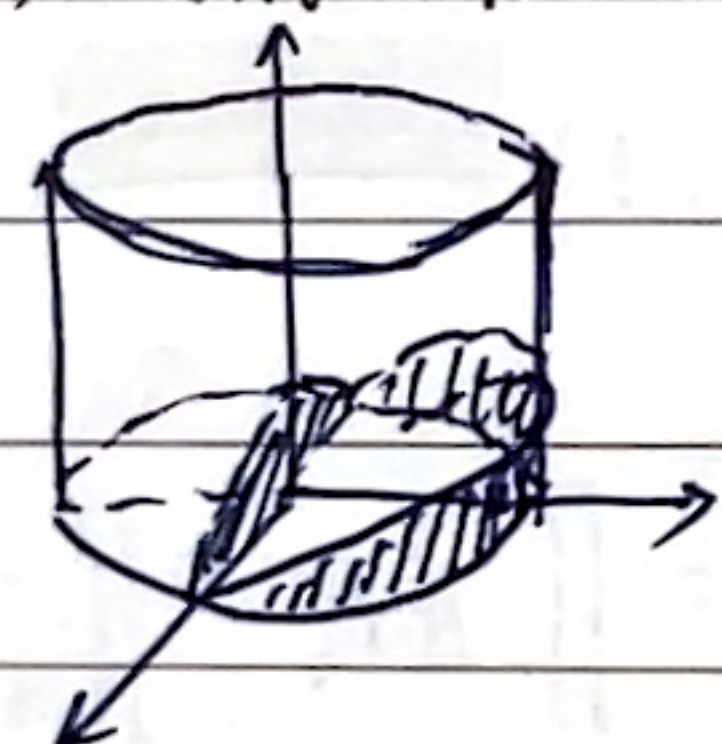
$A = \iint_{D_{xy}} \sqrt{1+z_x^2+z_y^2} dx dy \quad (D_{xy}: x^2+y^2 \leq 2x)$

$= \sqrt{2} \pi \cdot 1^2 = \sqrt{2} \pi.$

[例3] 求柱面 $x^2+y^2=R^2 (z \geq 0)$ 含在平面 $z=0$ 和 $z=y (y \geq 0)$ 间的面积.

所求部分柱面 S 方程: $y = \sqrt{R^2 - x^2}$

$\begin{cases} x^2+y^2=R^2 \\ z=y \end{cases} \Rightarrow x^2+z^2=R^2$



$\therefore S$ 在 xOz 平面投影 $D_{xz}: x^2+z^2=R^2 (z \geq 0)$

$\therefore A = \iint_{D_{xz}} \sqrt{1+z_x^2+z_y^2} dx dz = \iint_{D_{xz}} \frac{R}{\sqrt{R^2-x^2}} dx dz = R \int_{-R}^R \frac{1}{\sqrt{R^2-x^2}} dx \int_0^{\sqrt{R^2-x^2}} dz$

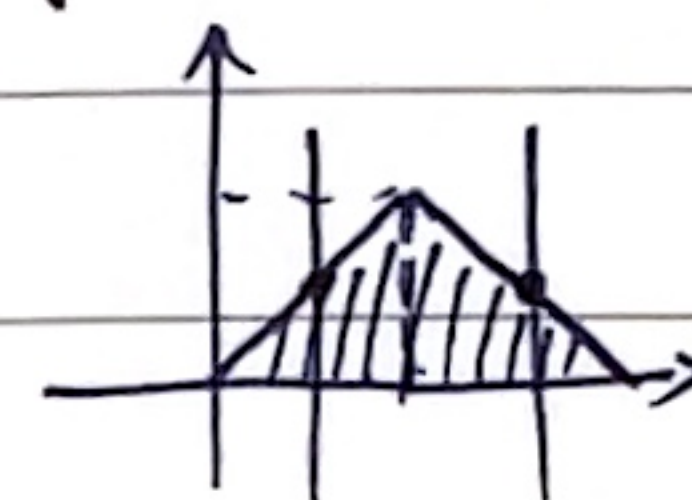
$= R \int_{-R}^R 1 dx = 2R^2$

章节复习.

1. 交换积分次序

(1) $\int_0^1 dy \int_y^{2-y} f(x,y) dx$

解:

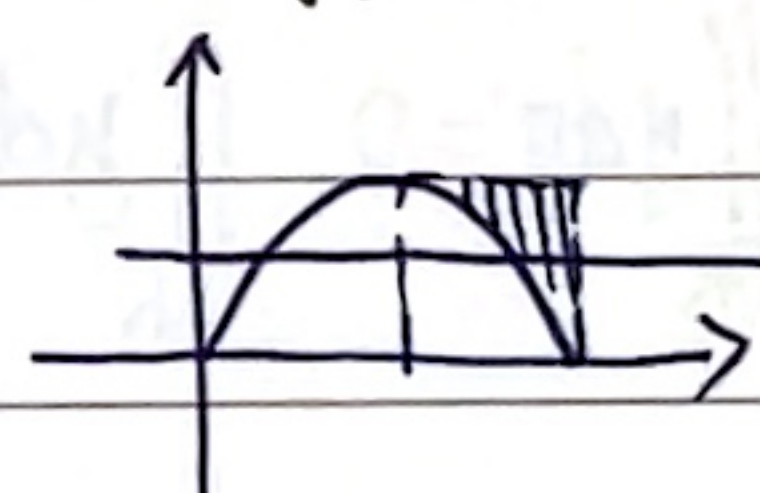


$D_{左}: 0 \leq x \leq 1, 0 \leq y \leq x$

$D_{右}: 1 \leq x \leq 2, 0 \leq y \leq 2-x$

$\therefore I = \int_0^1 dx \int_0^x f(x,y) dy + \int_1^2 dx \int_0^{2-x} f(x,y) dy$

(2) $\int_{\frac{\pi}{2}}^{\pi} dx \int_{\sin x}^1 f(x,y) dy$



只有 $y=\sin x \quad x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ 其反函数为 $x=\arcsin y$

$D: 0 \leq y \leq 1, \pi - \arcsin y \leq x \leq \pi$

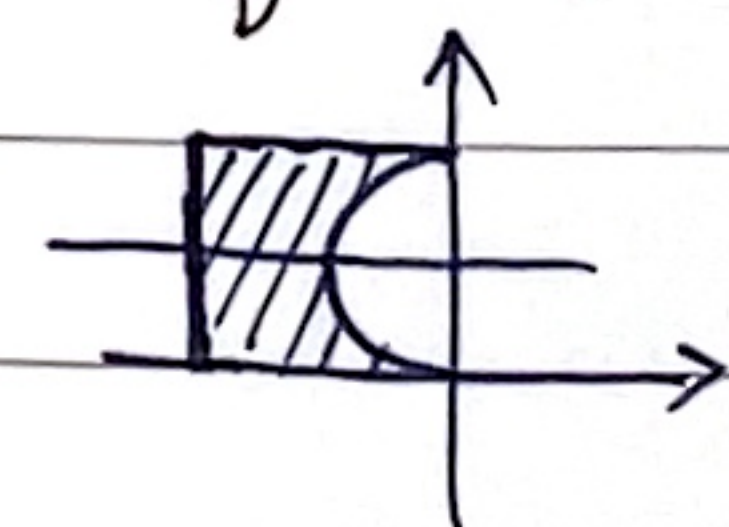
当 $x \in [\frac{\pi}{2}, \pi]$ 时, $\pi - x \in [0, \frac{\pi}{2}]$ 且 $y = \sin x = \sin(\pi - x)$

$\therefore \pi - x = \arcsin y \Rightarrow x = \pi - \arcsin y$

2. 平面区域 $D: x^2+y^2 \leq 1$, 则 $\iint_D (x^2-y^2) d\sigma = 0$.

$\because D$ 关于 $y=x$ 对称 $\therefore \iint_D x^2 d\sigma = \iint_D y^2 d\sigma$

3. $\iint_D y dx dy$, D 由直线 $x=-2, y=0, y=2$ 及曲线 $x=-\sqrt{2y-y^2}$ 围成.



$0 \leq y \leq 2, -2 \leq x \leq -\sqrt{1-(y-1)^2}$

$\int_0^2 y dy \int_{-2}^{-\sqrt{1-(y-1)^2}} dx = \int_0^2 y (2 - \sqrt{1-(y-1)^2}) dy = y^2 \Big|_0^2 - \int_0^2 y \sqrt{1-(y-1)^2} dy$

$\frac{y-1=\sin t}{t \in [-\frac{\pi}{2}, \frac{\pi}{2}]} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin t + 1) \cdot (-\cos t) \cos t dt + 4$

$= - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin t \cos^2 t + \cos^2 t) dt + 4$

$= 4 - 2x \frac{\pi}{2} \times \frac{1}{2} = 4 - \frac{\pi}{2}$

$\text{令 } t = \pi - \theta$

$= 4 - \frac{8}{3} \int_{\frac{\pi}{2}}^0 \sin^4 t (-dt)$

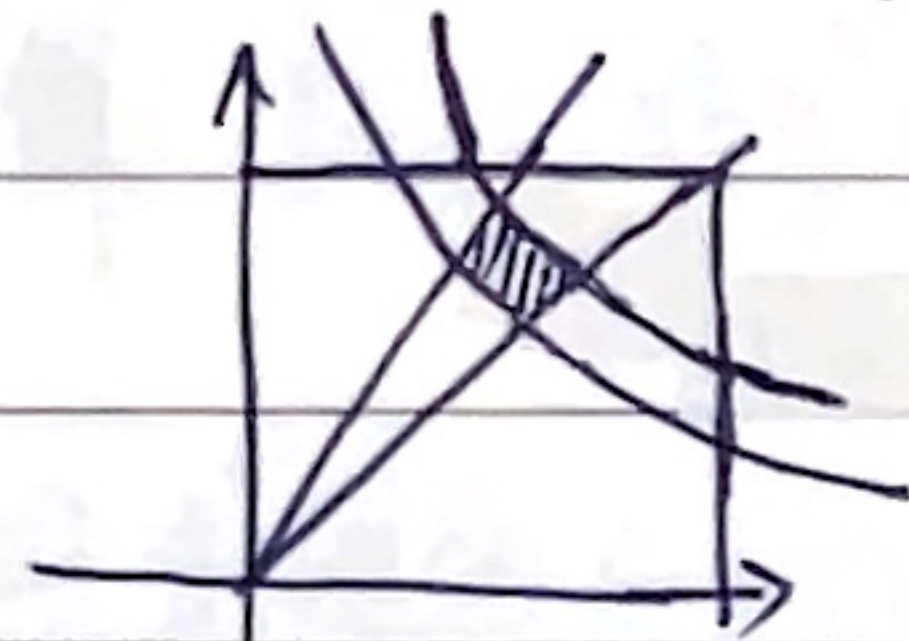
$= 4 - \frac{8}{3} \times \frac{3 \cdot 1}{4 \cdot 2} \frac{\pi}{2}$

Somewhere, something incredible,

$\iint_D y d\sigma = \iint_{[-2,0] \times [0,2]} y d\sigma - \iint_{\text{半圆}} y d\sigma = \int_{-2}^0 dx \int_0^2 y dy - \int_{\frac{\pi}{2}}^0 d\theta \int_0^{2\sin\theta} \sin\theta \cdot r dr = 4 - \frac{8}{3} \int_{\frac{\pi}{2}}^0 \sin^4 \theta d\theta$

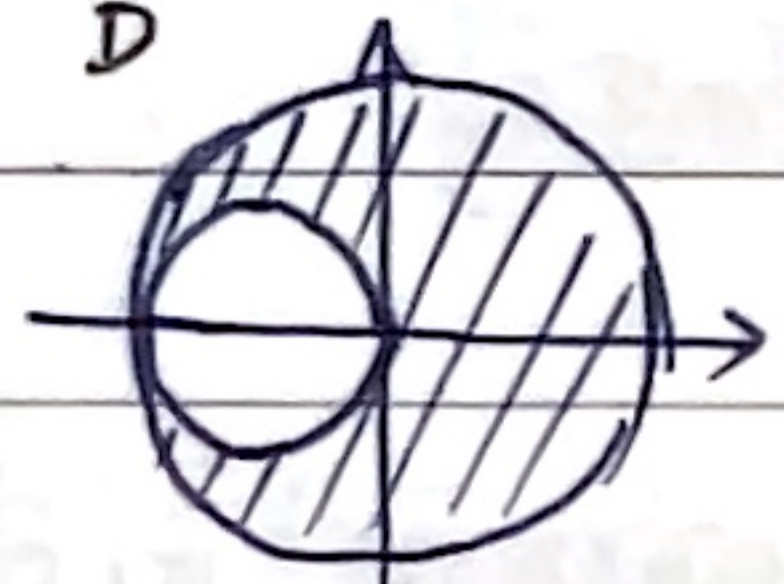
is waiting to be found.

4. D 由 $2xy=1$, $4xy=1$, $y=x$, $y=\sqrt{3}x$ 围成. 用极坐标形式表示 $\iint_D f(x,y) d\sigma =$ _____



$$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} d\theta \int_{\frac{1}{\sqrt{2}\sin\theta\cos\theta}}^{\frac{1}{\sqrt{4}\sin\theta\cos\theta}} f(r\cos\theta, r\sin\theta) r dr$$

5. $\iint_D (\sqrt{x^2+y^2}+y) d\sigma$, D 由 $x^2+y^2=4$, $(x+1)^2+y^2=1$ 围成.



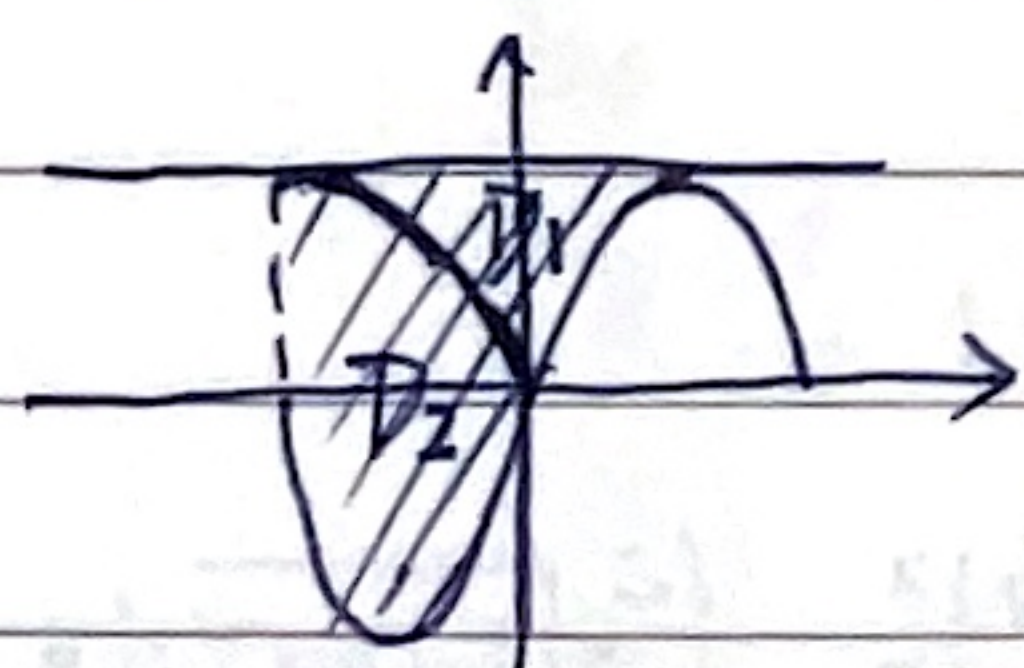
$$D = D_A - D_B$$

$\{D_A \text{ 关于 } x \text{ 轴对称, } y \text{ 关于 } y \text{ 为奇函数.} \therefore \iint_{D_A} y d\sigma = 0, \iint_{D_B} y d\sigma = 0.$

$$I = \iint_{D_A} (\sqrt{x^2+y^2}+y) d\sigma - \iint_{D_B} (\sqrt{x^2+y^2}+y) d\sigma = \int_0^{2\pi} d\theta \int_0^2 r \cdot r dr - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\theta \int_0^1 r \cdot r dr$$

$$= \frac{16}{3}\pi - \frac{32}{9}$$

6. D 由 $y=\sin x$, $x=\pm\frac{\pi}{2}$, $y=1$ 围成. 求 $\iint_D (xy^5-1) dx dy$



D_1 关于 y 轴对称, xy^5 关于 x 奇.

D_2 关于 x 轴对称, xy^5 关于 y 奇.

$$\text{原式} = - \iint_D dx dy = - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 - \sin x) dx = -\pi$$

7. $D: x^2+y^2 \leq 4, x \geq 0, y \geq 0, f(x) > 0$, 求 $I = \iint_D \frac{af(x)+b\sqrt{f(y)}}{\sqrt{f(x)}+\sqrt{f(y)}} d\sigma$

轮换对称性

$$\Rightarrow 2I = \iint a b + \iint b a = \iint (a+b) d\sigma = \pi(a+b)$$

$$\Rightarrow I = \frac{\pi}{2}(a+b)$$

8. 计算: $I = \iint_D \frac{(x+1)\sin\pi(x^2+y^2)}{x+y+z} d\sigma$, 其中 $D = \{(x,y) | 1 \leq x^2+y^2 \leq 4, x \geq 0, y \geq 0\}$

$$2I = \iint_D \sin\pi(x^2+y^2) d\sigma = \int_0^{\frac{\pi}{2}} d\theta \int_1^2 (\sin\pi r^2) \cdot r dr = \frac{\pi}{2} (\cos\pi r^2) \cdot \frac{1}{2\pi} \Big|_1^2 = -\frac{1}{2}$$

$$\Rightarrow I = -\frac{1}{4}$$

9. 曲面 $z=x^2+y^2$, $D: x^2+y^2 \leq 1$, 求其表面积.

$$A = \iint_{D_{xy}} \sqrt{1+z_x^2+z_y^2} dx dy = \iint_{D_{xy}} \sqrt{1+4x^2+4y^2} dx dy = \int_0^{2\pi} d\theta \int_0^1 \sqrt{1+4r^2} \cdot r dr = \frac{\pi}{6} - 5^{3/2}$$